LEDGER · 2026-09-01 · MATERIALS SCIENCE / CHEMICAL ENGINEERING

Inositol Hexaphosphate (IP6) Substrate Passivation via Ambient Immersion

PRODUCT

A product dossier of The Absence Audit. The current audit flags this concept — Below the 2x step-change bar — and the full disclosure ships inside the dossier.

Incumbent benchmark	Henkel BONDERITE M-ZN 4333
Entry application	ambient-temperature pre-treatment of Q235 carbon steel and magnesium alloys in automotive aftermarket and white-goods manufacturing

Measured advantage	ASTM B117 Neutral Salt Spray resistance on bare steel: ~72-120 hours for silane-doped IP6 vs 120+ hours for Henkel BONDERITE M-ZN 4333 (<1.0x delta)
Time to first revenue	3 months
Gross margin at parity	93.8%
Startup CapEx	$41,000
Payback from first sale	0.0 months
Capital productivity	625.89x
Regulatory risk	Low — no material regulatory CapEx reported ($0 estimated)

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Mechanism, operating protocol with real conditions and yields, computed unit economics with a six-scenario stress test, the absence audit, and every resolved citation.

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Assessed 2026-09-01 against seven gates plus the voiding sub-tests. How the method works · Browse all concepts published scientific literature and have not been commercially validated; several involve chemical, biological or regulated processes that require appropriate expertise, facilities, permits and safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.